

String and string methods:

String

A **string** is a collection of one or more characters.

Strings can be written using:

- Single quotes ' '
- Double quotes " "
- Triple quotes ''' ''' or """ """

For multiline strings, triple quotes are used.

Example

```
str1 = "Abhi"
```

```
print(str1)
```

Multiline String

```
message = """
```

```
This is a  
multiline string  
"""
```

```
print(message)
```

The uploaded PDF introduces strings and then covers common string methods with examples.

2. Creating a String

A string can be created by assigning text to a variable.

```
str1 = "Abhi"
```

```
print(str1)
```

String Methods

The PDF covers the following string methods:

1. lower()
2. capitalize()
3. upper()
4. title()
5. strip()
6. lstrip()
- 7.rstrip()
8. replace()
9. split()
10. join()
11. find()
12. index()
13. startswith()
14. endswith()
15. isdigit()
16. isalpha()
17. isalnum()
18. isnumeric()

- 19. isspace()
 - 20. swapcase()
 - 21. zfill()
 - 22. rjust()
 - 23. ljust()
 - 24. center()
-

3. lower()

The lower() method converts a string into lowercase.

Example

```
string1 = "Python is Great"
```

```
print(string1.lower())
```

Output

```
python is great
```

4. capitalize()

The capitalize() method converts the **first character** of the string to uppercase and the remaining characters to lowercase.

Example

```
string = "python is great"
```

```
print(string.capitalize())
```

Output

```
Python is great
```

5. upper()

The upper() method converts a string into uppercase.

Example

```
str1 = "Python is great"
```

```
print(str1.upper())
```

Output

```
PYTHON IS GREAT
```

6. title()

The title() method converts the first character of **each word** to uppercase.

Example

```
str1 = "python is great"
```

```
print(str1.title())
```

Output

```
Python Is Great
```

7. strip()

The strip() method returns a **trimmed version** of the string by removing spaces from both the beginning and the end.

Example

```
string = " Python is great "
```

```
print(string.strip())
```

Output

```
Python is great
```

8. lstrip()

The lstrip() method removes whitespace from the **left side** of the string.

Example

```
str1 = " Python is great"
```

```
print(str1.lstrip())
```

Output

```
Python is great
```

9. rstrip()

The rstrip() method removes whitespace from the **right side** of the string.

Example

```
str1 = "Python is great "
```

```
print(str1.rstrip())
```

Output

```
Python is great
```

10. replace()

The replace() method returns a string where a specified value is replaced with another specified value.

Syntax

```
string.replace(old, new)
```

Example

```
name = "abhi"
```

```
print(name.replace("a", "A"))
```

Output

```
Abhi
```

11. split()

The split() method splits a string at the specified separator and returns a list.

Example

```
string = "This is a string"
```

```
print(string.split(" "))
```

Output

```
['This', 'is', 'a', 'string']
```

To find the number of elements:

```
print(len(string.split(" ")))
```

12. join()

The join() method joins the elements of an iterable into a string.

Example

```
string = "This is a string"
```

```
print("-".join(string.split(" ")))
```

Output

```
This-is-a-string
```

13. find()

The find() method searches the string for a specified value and returns the position where it is found. It returns the position of the **first occurrence**.

Example

```
string = "This is a string"
```

```
print(string.find("is"))
```

Output

```
2
```

The first "is" starts at index 2.

14. index()

The index() method searches the string for a specified value and returns the position where it is found.

Example

```
string = "String"
```

```
print(string.index("S"))
```

Output

```
0
```

15. startswith()

The startswith() method returns True if the string starts with the specified value.

Example

```
text = "Hello, welcome to my world"
```

```
x = text.startswith("Hello")
```

```
print(x)
```

Output

```
True
```

16. endswith()

The endswith() method returns True if the string ends with the specified value.

Example

```
name = "Abhi"
```

```
print(name.endswith("i"))
```

Output

```
True
```

17. isdigit()

The isdigit() method returns True if all characters in the string are digits.

Example

```
password = "secret_123"
```

```
for i in password:
    if i.isdigit():
        print("valid", i)
```

For example:

```
text = "12345"
```

```
print(text.isdigit())
```

Output:

True

18. isalpha()

The isalpha() method returns True if all characters in the string are alphabet letters (a-z).

Example

```
password = "secret"
```

```
for i in password:
    if i.isalpha():
        print("valid", i)
```

Another example:

```
text = "Python"
```

```
print(text.isalpha())
```

Output:

True

19. isalnum()

The isalnum() method returns True if all characters are a combination of alphabetic and numeric characters.

Example

```
text = "12345abc"
```

```
print(text.isalnum())
```

Output

True

A string containing spaces or special characters will return False.

20. isnumeric()

The isnumeric() method checks whether all characters in the string are numeric.

Example

```
text = "12345"
```

```
print(text.isnumeric())
```

Output

True

21. isspace()

The isspace() method returns True if all characters in the string are whitespace characters.

Example

```
text = "  "
```

```
x = text.isspace()
```

```
print(x)
```

Output

True

22. swapcase()

The swapcase() method converts:

- Lowercase characters → Uppercase
- Uppercase characters → Lowercase

Example

```
name = "Abhi"
```

```
print(name.swapcase())
```

Output

aBHI

23. zfill()

The zfill() method fills the string with zeros at the beginning until it reaches the specified width.

Example

```
text = "50"
```

```
x = text.zfill(10)
```

```
print(x)
```

Output

0000000050

24. rjust()

The rjust() method returns a **right-justified** version of a string.

Example

```
text = "banana"
```

```
x = text.rjust(20)
```

```
print(x)
```

The string is shifted toward the right within a field of width 20.

25. ljust()

The ljust() method returns a **left-justified** version of a string.

Example

```
text = "banana"
```

```
x = text.ljust(10)
```

```
print(x)
```

The string is positioned toward the left within a field of width 10.

26. center()

The center() method returns a **centered version** of a string.

Example

```
text = "banana"
```

```
x = text.center(10)
```

```
print(x)
```

Output

```
banana
```

The string is positioned in the center of the specified width.

27. Example – Character-by-Character String Processing

The PDF also contains an example where a string is processed character by character.

```
string = "Python"
```

```
res = ""
```

```
for i in string:
```

```
    if i >= "i":
```

```
        res = res + i.lower()
```

```
    else:
```

```
        res = res + i.upper()
```

```
print(res)
```

This example demonstrates how string methods such as upper() and lower() can be used inside a for loop to process individual characters.